

CODE:

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import numpy as np
import random
grid_size = 4
start = (0, 0)
goal = (3, 3)
actions = { 0: (-1, 0),
            1: (1, 0),
            2: (0, -1),
            3: (0, 1) }
Q = np.zeros((grid_size, grid_size, 4))
alpha = 0.1
gamma = 0.9
epsilon = 0.2
for episode in range(1000):
    state = start
    for step in range(50):
        r, c = state
        if random.random() < epsilon:
            action = random.randint(0, 3)
        else:
            action = np.argmax(Q[r, c])
        dr, dc = actions[action]
        nr, nc = r + dr, c + dc
        if nr < 0 or nr >= grid_size or nc < 0 or nc >= grid_size:
            nr, nc = r, c
            reward = -5
        elif (nr, nc) == goal:
            reward = 10
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    else:
        reward = -1
        old = Q[r, c, action]
        future = np.max(Q[nr, nc])
        Q[r, c, action] = old + alpha * (reward + gamma * future - old)
        state = (nr, nc)
        if state == goal:
            break
state = start
path = [state]
for step in range(20):
    r, c = state
    action = np.argmax(Q[r, c])
    dr, dc = actions[action]
    state = (max(0, min(grid_size - 1, r + dr)),
             max(0, min(grid_size - 1, c + dc)))
    path.append(state)
    if state == goal:
        break
print("Start:", start)
print("Goal:", goal)
print("Learned Path:", path)
print("Goal Reached:", state == goal)

```

OUTPUT:

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Start: (0, 0)
Goal: (3, 3)
Learned Path: [(0, 0), (1, 0), (1, 1), (2, 1), (2, 2), (3, 2), (3, 3)]
Goal Reached: True

```